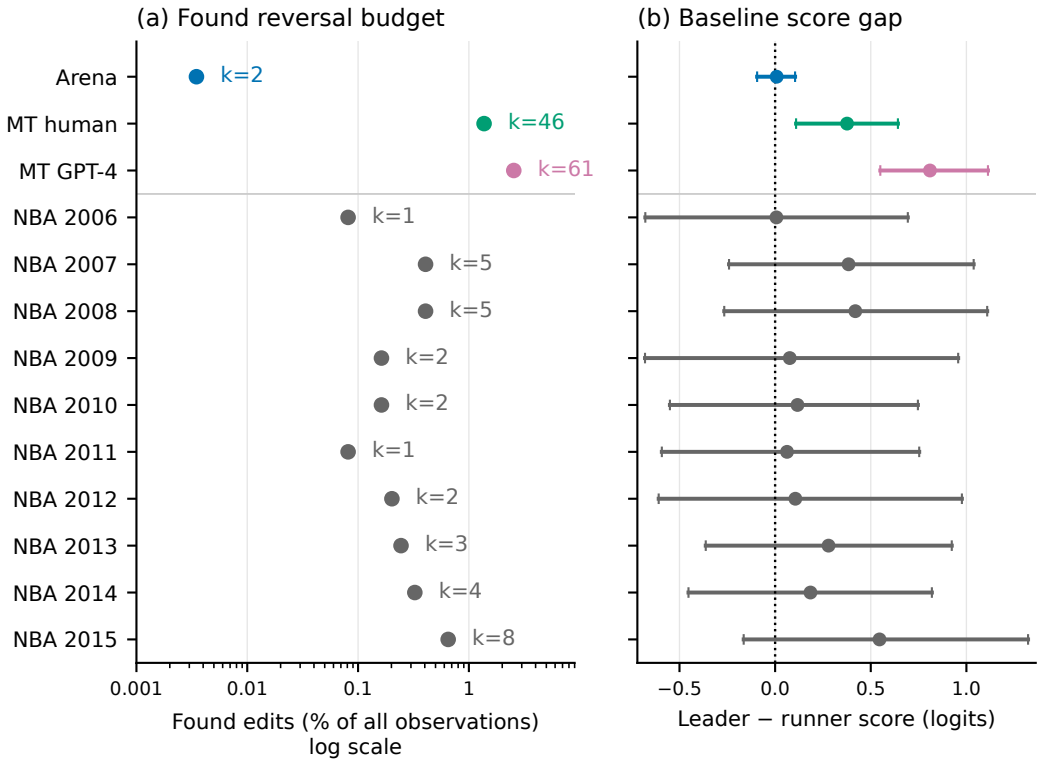
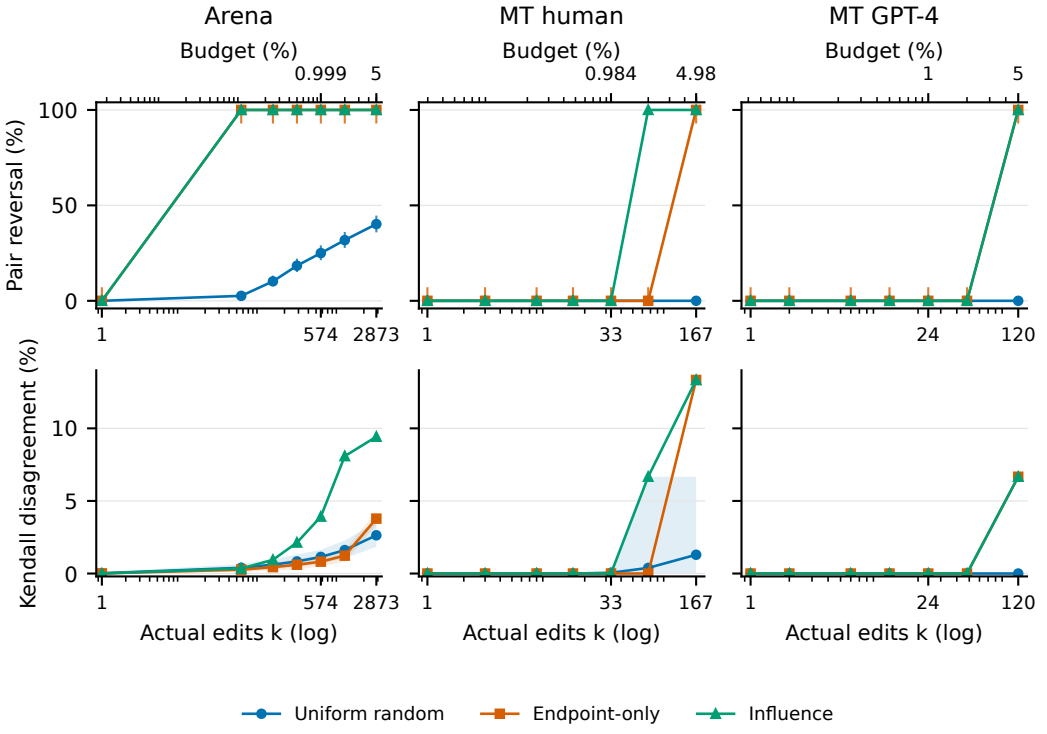
Primary stability



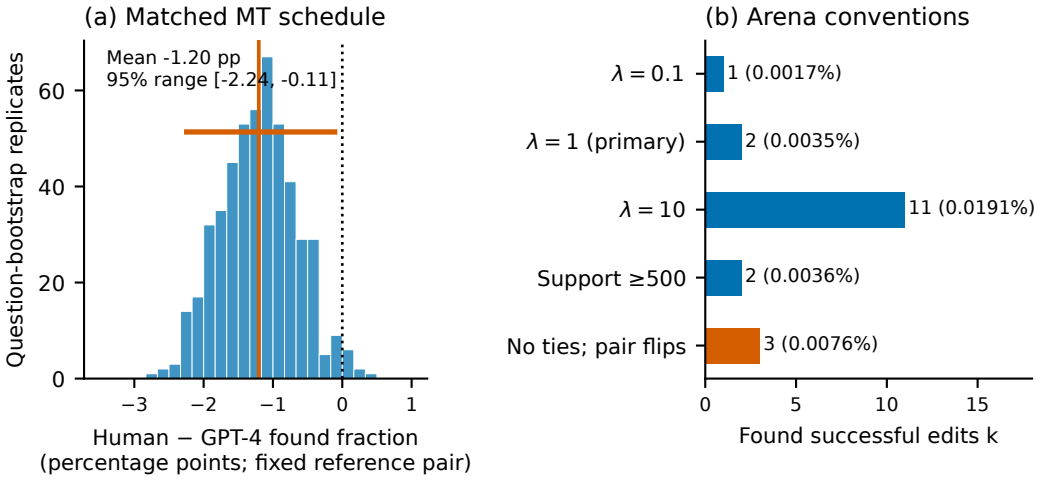
Small found attack budgets coincide with unresolved baseline leading contrasts in these fits. (a) Successful influence-prefix attacks on each original leading pair, shown as a percentage of all observations with absolute edit counts. These are upper bounds on the smallest successful attack, not certified minima. (b) Unperturbed score gaps and 95% percentile ranges from 500 cluster-bootstrap replicates, with the original pair held fixed. Clusters are Arena battles, MT questions, and NBA dates. Ranges are conditional descriptive uncertainty, without simultaneous or post-selection coverage guarantees. NBA seasons are separate fits; scores are in logit units.

Strategy curves



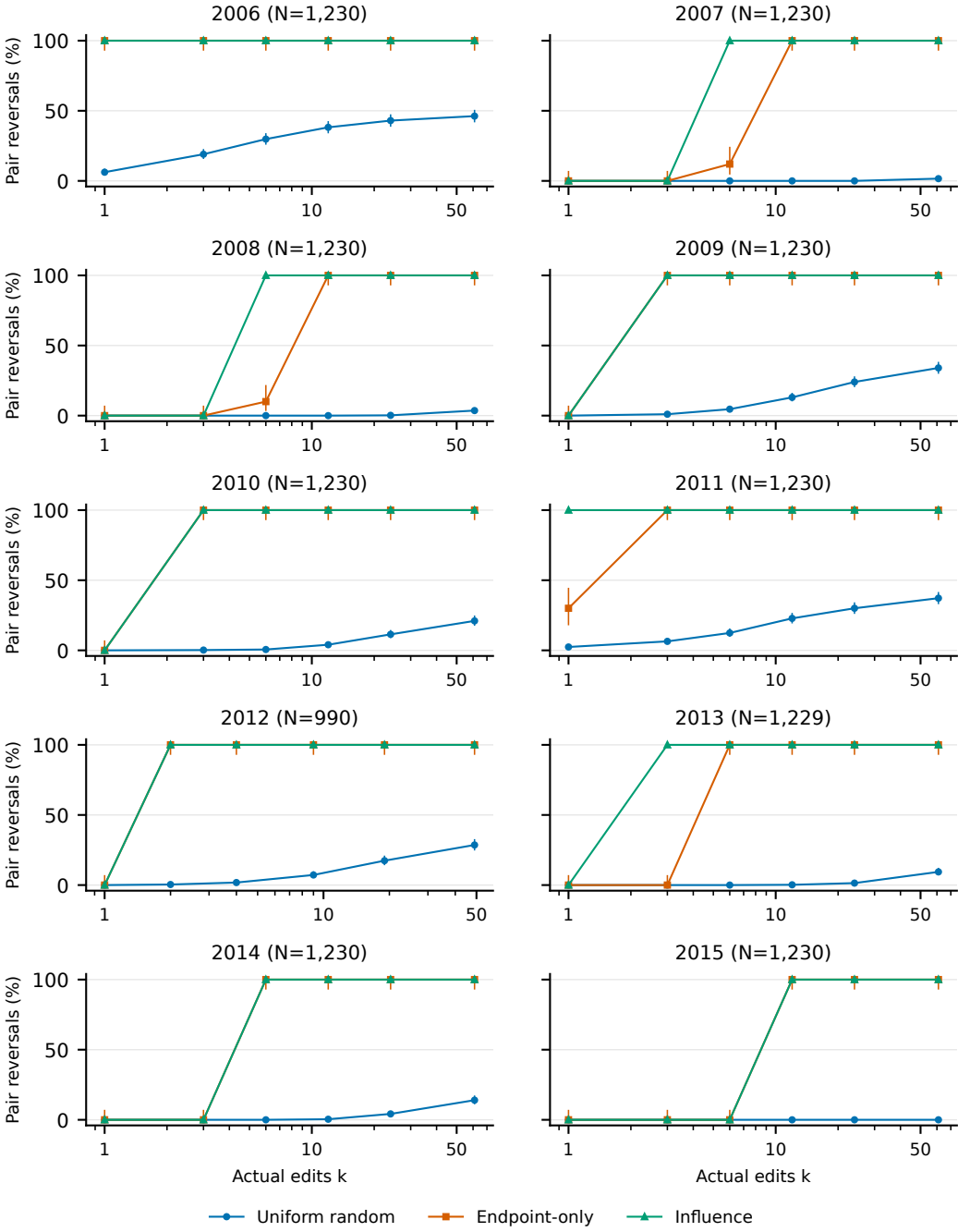
Targeting reverses leading contrasts more often than random edits while producing a different degree of whole-ranking change. The upper row shows fixed original leader–runner reversal rates; vertical bars are 95% exact binomial Monte Carlo intervals across 500 uniform-random or 50 endpoint-only runs. Deterministic influence attacks have no Monte Carlo interval. The lower row shows mean normalized Kendall pairwise disagreement; shading gives empirical 2.5–97.5 percentiles across perturbation runs, not data-bootstrap uncertainty. Markers are measured integer budgets; connecting lines guide the eye. Top axes convert counts to percentages of all observations. Every displayed strategy uses its full budget; exact budget mappings appear in the appendix. Reversing the original leading pair displaces the original winner, but the original runner-up need not become the new winner. A winner change can also occur without reversal of that pair.

Controls



The matched MT contrast persists on shared support, while Arena counts depend on fitting conventions. (a) Human minus GPT-4 found attack fractions on the matched MT schedule, holding the GPT-4–Claude-v1 reference pair fixed. The histogram contains 500 paired question-bootstrap replicates with human-label resampling; orange marks show the mean and 95% percentile range (percentage points). This reference-pair analysis is a post-hoc follow-up; five human resamples already reverse the pair and require zero additional edits. (b) Arena found counts under changed conventions; labels give percentages using each fit's own observation count. Removing ties swaps the initial leader and runner, so its attack addresses the newly selected contrast. These controls describe the observed snapshots and heuristic, without identifying a causal evaluation-design effect or a globally minimal attack budget.

NBA curves



All ten NBA seasons under the same exact-refit perturbation protocol. Markers show actual integer edit counts on logarithmic axes; lines guide the eye between measured budgets. Vertical bars show 95% exact binomial Monte Carlo intervals for uniform random (500 runs) and endpoint-only (50 runs); the influence sequence is deterministic. Seasons have distinct original leading pairs and sample sizes. Reversal of the original leading pair implies that the original leader loses first place, but the new winner can be a third team. Winner change can also occur without reversal of that pair. Fractional budget mappings are reported separately.

Primary evidence

Ranking	N	M	Decisive	Found k	$100k/N$ (%)	Gap [95% range]
Arena	57,477	64	39,716	2	0.0035	0.008 [-0.094, 0.104]
MT human	3,355	6	2,575	46	1.3711	0.376 [0.110, 0.642]
MT GPT-4	2,400	6	1,800	61	2.5417	0.809 [0.550, 1.113]
NBA 2006	1,230	30	1,230	1	0.0813	0.007 [-0.679, 0.694]
NBA 2007	1,230	30	1,230	5	0.4065	0.384 [-0.241, 1.038]
NBA 2008	1,230	30	1,230	5	0.4065	0.419 [-0.266, 1.109]
NBA 2009	1,230	30	1,230	2	0.1626	0.076 [-0.680, 0.957]
NBA 2010	1,230	30	1,230	2	0.1626	0.117 [-0.550, 0.746]
NBA 2011	1,230	30	1,230	1	0.0813	0.062 [-0.592, 0.753]
NBA 2012	990	30	990	2	0.2020	0.106 [-0.609, 0.976]
NBA 2013	1,229	30	1,229	3	0.2441	0.279 [-0.363, 0.924]
NBA 2014	1,230	30	1,230	4	0.3252	0.185 [-0.453, 0.820]
NBA 2015	1,230	30	1,230	8	0.6504	0.546 [-0.164, 1.322]

Primary fixed-snapshot evidence. N counts all comparisons, including ties; M counts entrants. Found k is a successful influence-prefix attack and only an upper bound. Each gap is the original leader minus runner in logit units; brackets give the conditional 95% cluster-bootstrap percentile range (500 replicates).

Datasets

Ranking	Original leader > runner	N	Ties (%)	Min support	Clusters
Arena	GPT-4-1106 > GPT-4-0125	57,477	30.9	100	57,477
MT human	GPT-4 > Claude-v1	3,355	23.2	996	80
MT GPT-4	GPT-4 > Claude-v1	2,400	25.0	800	80
NBA 2006	SAS > DET	1,230	0.0	82	164
NBA 2007	DAL > PHO	1,230	0.0	82	163
NBA 2008	BOS > LAL	1,230	0.0	82	162
NBA 2009	CLE > LAL	1,230	0.0	82	163
NBA 2010	CLE > ORL	1,230	0.0	82	164
NBA 2011	SAS > CHI	1,230	0.0	82	164
NBA 2012	SAS > CHI	990	0.0	66	120
NBA 2013	MIA > OKC	1,229	0.0	81	163
NBA 2014	SAS > OKC	1,230	0.0	82	163
NBA 2015	GSW > ATL	1,230	0.0	82	162

Dataset coverage and original selected pairs. Arena names abbreviate `gpt-4-1106-preview` and `gpt-4-0125-preview`; MT uses `gpt-4` and `claude-v1`. Minimum support counts incident comparisons per entrant. Bootstrap clusters are battles for Arena, questions for MT, and dates for NBA. NBA labels are season-ending years.

One-percent budgets

Ranking	k	Random (%)	Endpoint (%)	Influence (%)	Winner (%)	Kendall (%)
Arena	574	25.0	100.0	100.0	100.0	3.92
MT human	33	0.0	0.0	0.0	0.0	0.00
MT GPT-4	24	0.0	0.0	0.0	0.0	0.00
NBA 2006	12	38.2	100.0	100.0	100.0	2.99
NBA 2007	12	0.0	100.0	100.0	100.0	2.53
NBA 2008	12	0.0	100.0	100.0	100.0	3.22
NBA 2009	12	13.0	100.0	100.0	100.0	2.30
NBA 2010	12	4.0	100.0	100.0	100.0	4.60
NBA 2011	12	22.8	100.0	100.0	100.0	2.30
NBA 2012	9	7.2	100.0	100.0	100.0	3.68
NBA 2013	12	0.2	100.0	100.0	100.0	3.22
NBA 2014	12	0.4	100.0	100.0	100.0	2.53
NBA 2015	12	0.0	100.0	100.0	100.0	2.53

Effects at $k = \lfloor 0.01N \rfloor$ actual edits. Random, Endpoint and Influence are fixed-pair reversal percentages; random and endpoint rates average 500 and 50 runs respectively. Winner is any change of global winner under the deterministic influence attack. Kendall is its normalized pairwise disagreement, expressed as a percentage. Rounding makes actual fractions slightly below 1% in some datasets; exact mappings are supplied separately.

Convention sensitivity

Convention	N	M	Found k	$100k/N$ (%)
Arena: $\lambda = 0.1$	57,477	64	1	0.0017
Arena: $\lambda = 1$	57,477	64	2	0.0035
Arena: $\lambda = 10$	57,477	64	11	0.0191
Arena: support ≥ 500	54,985	55	2	0.0036
Arena: no ties (pair flips)	39,716	64	3	0.0076
MT human: $\lambda = 0.1$	3,355	6	46	1.3711
MT human: $\lambda = 1$	3,355	6	46	1.3711
MT human: $\lambda = 10$	3,355	6	46	1.3711
MT human: expert only	2,668	6	41	1.5367
MT GPT-4: $\lambda = 0.1$	2,400	6	61	2.5417
MT GPT-4: $\lambda = 1$	2,400	6	61	2.5417
MT GPT-4: $\lambda = 10$	2,400	6	61	2.5417
MT GPT-4: consistent only	2,020	6	47	2.3267

Convention sensitivity of found influence-prefix budgets. Ties enter the primary fit as half-wins. Regularization uses the summed likelihood plus $\lambda\|\theta\|^2/2$. Arena no-tie filtering swaps the initial leading pair; the attack is redefined on that fit's selected pair. Arena support filtering and MT filters also change N, which remains the denominator. Counts do not certify minimum edits or robustness below the found count.

Matched controls

Matched statistic / resampling	Runs	Mean difference (pp)	95% range (pp)
Reselected pair / human labels	100	-1.15	[-1.38, -0.88]
Reselected pair / joint questions	500	-1.24	[-2.28, -0.11]
Fixed reference pair / joint questions	500	-1.20	[-2.24, -0.11]

Paired human minus GPT-4 found attack fractions on common MT support (percentage points). The first row varies human labels over 100 selections on 1,814 fixed cells; its range captures label selection, not question uncertainty. The next two rows use 500 joint question-bootstrap draws with matched counts and human-label resampling. Reselected-pair statistics choose each resample's top two; the post-hoc fixed-reference statistic uses GPT-4–Claude-v1 and assigns zero edits to already reversed contrasts. Ranges are empirical percentiles of heuristic statistics, not coverage guarantees for minimum budgets.

Integer budget map

Ranking	Actual edits k (percentage of all observations)
Arena	1 (0.0017%); 57 (0.0992%); 143 (0.2488%); 287 (0.4993%); 574 (0.9987%); 1149 (1.9991%); 2873 (4.9985%)
MT human	1 (0.0298%); 3 (0.0894%); 8 (0.2385%); 16 (0.4769%); 33 (0.9836%); 67 (1.9970%); 167 (4.9776%)
MT GPT-4	1 (0.0417%); 2 (0.0833%); 6 (0.2500%); 12 (0.5000%); 24 (1.0000%); 48 (2.0000%); 120 (5.0000%)
NBA 2006	1 (0.0813%); 3 (0.2439%); 6 (0.4878%); 12 (0.9756%); 24 (1.9512%); 61 (4.9593%)
NBA 2007	1 (0.0813%); 3 (0.2439%); 6 (0.4878%); 12 (0.9756%); 24 (1.9512%); 61 (4.9593%)
NBA 2008	1 (0.0813%); 3 (0.2439%); 6 (0.4878%); 12 (0.9756%); 24 (1.9512%); 61 (4.9593%)
NBA 2009	1 (0.0813%); 3 (0.2439%); 6 (0.4878%); 12 (0.9756%); 24 (1.9512%); 61 (4.9593%)
NBA 2010	1 (0.0813%); 3 (0.2439%); 6 (0.4878%); 12 (0.9756%); 24 (1.9512%); 61 (4.9593%)
NBA 2011	1 (0.0813%); 3 (0.2439%); 6 (0.4878%); 12 (0.9756%); 24 (1.9512%); 61 (4.9593%)
NBA 2012	1 (0.1010%); 2 (0.2020%); 4 (0.4040%); 9 (0.9091%); 19 (1.9192%); 49 (4.9495%)
NBA 2013	1 (0.0814%); 3 (0.2441%); 6 (0.4882%); 12 (0.9764%); 24 (1.9528%); 61 (4.9634%)
NBA 2014	1 (0.0813%); 3 (0.2439%); 6 (0.4878%); 12 (0.9756%); 24 (1.9512%); 61 (4.9593%)
NBA 2015	1 (0.0813%); 3 (0.2439%); 6 (0.4878%); 12 (0.9756%); 24 (1.9512%); 61 (4.9593%)

Exact integer budgets and their percentages of all observations. Nominal budgets are one edit and $\max(1, \lfloor fN \rfloor)$ for $f \in \{0.001, 0.0025, 0.005, 0.01, 0.02, 0.05\}$. Duplicate budgets are removed. For every primary dataset, strategy and replicate, actual edits equal the integer budget: no displayed primary curve saturates.